

OCEAN ACIDIFICATION

The continued decrease in pH of the ocean, which is primarily caused by the increased amount of CO₂ production by humans.

RESEARCH QUESTIONS

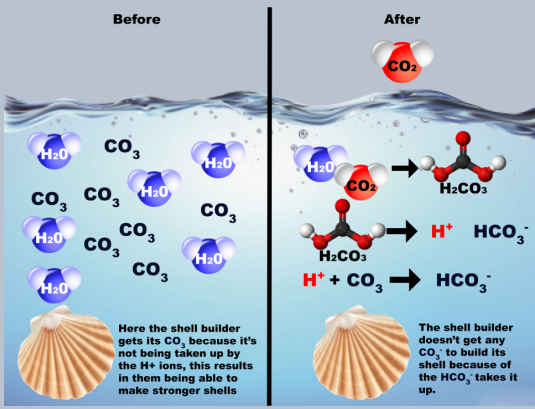
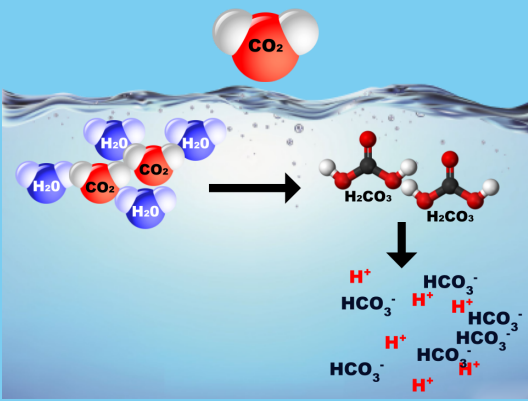
1. How much time does the acidity of the water increase given a variety of CO₂ being added to the system?
- a. Independent: Time

b. Dependent: pH change of water

c. Control: CO₂ added (multiple trials to achieve the variety), Amount/Condition of Water
2. How does CO₂ being added to water impact the conductivity of the solution?
- a. Independent: CO₂ added

b. Dependent: Conductivity

c. Control: Amount/Condition of water



WHAT IS THE PROBLEM?

The ocean's acidity, or pH, has lowered by 0.1 pH level. While that may not sound like much, because of the logarithmic nature of the pH scale, this translates to around a 30% rise in acidity of the ocean.

WHAT IS CAUSING THE PROBLEM CHEMICALLY?

Essentially what is happening is the increased amounts of CO₂ in the atmosphere is being absorbed by the ocean. This is a good thing usually because it keeps the CO₂ out of the atmosphere, acting as a large carbon sink. The problem with this, however, is that this added CO₂ is causing the ocean acidity to rise. This is happening because when the CO₂ reacts with the water, it results in an end product of H⁺ atoms as well as HCO₃ atoms. This means that the molarity of H⁺ is increasing, causing the acidification.

HOW CAN WE SOLVE THE PROBLEM?

There are a few different options to solving ocean acidification, to name a few: Cutting carbon emissions, geoengineering, and self reduction (The Ocean Portal Team, 2023). First, the most seemingly obvious solution of cutting carbon emissions. While this may sound ideal in theory it has a few drawbacks. While it may stop the acidification of the ocean from rising a lot more, it won't stop it right away. This is due to the time it takes for atmospheric carbon dioxide to be absorbed into the ocean. The next idea is geoengineering, which would theoretically allow us to maintain our current lifestyles. This sounds great, but requires releasing an organism that will help remove the carbon dioxide, namely phytoplankton. This sounds great, except when these phytoplankton die, they will fall down into the deep ocean, potentially raising the acidity down there. The final solution is cutting our usage of energy and being responsible with our usage of things created by releasing carbon dioxide. Unfortunately the chances of this working on mass is very low. People don't want to have to change because of something that feels so distant, therefore it's unlikely that this will work. So while these ideas may sound great in theory, they all have their drawbacks. The best path forward is a combination of all three, so then we can try to balance out all the pros and cons. This also makes it more likely that things will actually be accomplished.

